Shiyu Chen 陈时宇

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Research interests: autonomous exploration & active perception · semantic mapping & LiDAR–camera fusion · next-best-view planning · VLM-in-the-loop industrial inspection

EDUCATION

Carnegie Mellon University — M.S. in Mechanical Engineering (Research) Aug 2025 – May 2027 (expected)
Computational Engineering and Robotics Lab (CERLAB), advised by Prof. Kenji Shimada Pittsburgh, PA

Univ. of Cincinnati & Chongqing University — Joint Program, B.S. / B.Eng. in Mechanical Engineering 2020 – 2025
Robotics Minor (UC) · UC GPA 3.71/4.0 · CQU GPA 3.58/4.0 · Dean's List ×4 Cincinnati, OH / Chongqing, CN

RESEARCH EXPERIENCE

Graduate Researcher, CERLAB Roboteam (CMU × YKK factory-automation inspection) Oct 2025 – Present

5-DoF semantic exploration with VLM-guided active inspection **(co-first author · arXiv:2609.19460 · submitted to ICRA 2027)**

- Spot tilts its body (pitch, roll) to see the upper surfaces a level camera misses — **5-DoF viewpoints** on top of planar motion.
- Semantic mapping: per-instance detection masks back-projected into 3D object boxes in the occupancy map.
- Viewpoint sampler picks body postures by expected coverage gain; aim-aligned execution avoids needless reorientation.
- **VLM-assisted viewpoint pruning** from observation history + bird's-eye-view map; geometric fallback so the stack never blocks.
- Built the Isaac Sim benchmark (three industrial scenes, ground-truth surfaces, plug-in planners, Dockerized; C++/ROS planner).
- **+8–10 pp** coverage and **17–32% less exploration time** vs. the planar 3-DoF baseline; **53–73% fewer postures** than competing 5-DoF methods; highest mean object-coverage AUC.
- Demonstrated on a real Spot in a machine shop (five target machines).

Real-time LiDAR–camera semantic registration on a legged platform

- Fused Ouster LiDAR geometry with Insta360 X4 panoramic imagery on a Boston Dynamics Spot.
- **10 FPS** at full LiDAR rate, **<100 ms** latency, **<10 px** reprojection error, stable under gait vibration (3D-printed rigid sensor mount).

Research Assistant, 3D Reconstruction of Shoe Interiors for Footwear Fit (CERLAB) Sep 2026 – Present

- Sole researcher: geometry-only 3D reconstruction of the inside of a shoe for fit assessment (e.g., diabetic feet); scope extended to garment interiors.
- Designed a slender stereo endoscopic probe — two side-looking global-shutter cameras (**34 mm baseline, 148° FOV**), LED ring, printed wall fiducials; a cone-mirror laser-ring route is validated in parallel.
- Error budget: ≈0.3 mm depth error at 100 mm standoff from a 34 mm baseline; sharp range verified against lens depth of field. **Target reconstruction error ≤ 1 mm.**
- Designed the manual rail fixture for static point-by-point scans (push/pull + roll); test plan on enlarged 3D-printed cavity halves with CAD ground truth, then real shoes. Hardware procured; fixtures in print.

Undergraduate Researcher, Cable-Driven Soft Robotic Arm for Minimally Invasive Surgery Sep 2024 – May 2025
Univ. of Cincinnati, advised by Prof. Janet (Jiangxia) Dong

- Designed and prototyped a wire-actuated continuum arm: **0.27 mm/step** resolution, **>230°** bending range.
- Modeled nonlinear actuation; led hardware–control integration; published in *ASME J. Medical Diagnostics* (2026).

Undergraduate Researcher, Deep-Learning Surface-Defect Segmentation Jan 2022 – Apr 2022
Chongqing University, advised by Prof. Bo Yang

- Semantic segmentation of aluminium surface defects with SegNet + ResNet encoder on a 2,658-image dataset, **97.64%** accuracy (after an EfficientDet detection baseline); collected 1,125 carbon-fibre vibration-source samples.

PUBLICATIONS & PATENTS

X. Zhan*, **S. Chen***, K. Shimada, "Pose-aware Legged Robot Semantic Exploration with Omnidirectional Perception in Confined Unknown Environments," *arXiv:2609.19460*, Sep 2026. Submitted to ICRA 2027 (under review). *Equal contribution.

Y. Wang, C. Liu, K. Jiang, B. Wu, **S. Chen**, J. Dong, A. Ashfaq, "Innovative Flexible Robotic Arm for Enhanced Precision in Transaortic Surgical Myectomy," *ASME Journal of Medical Diagnostics*, 9(2): 021001, 2026.

L. Liu, **S. Chen**, Z. Tang, "Intelligent Diagnosis of Bearing Faults Based on SDP Image Fusion," *Int. Conf. on Image Processing, Computer Vision and Machine Learning (ICICML)*, Chengdu, 2023.

Patents (China, as co-inventor): 4 granted invention patents (2025–2026) on deep-learning machine vision for steelmaking process monitoring (steel-stream landing-point, slag-splashing and slag-discharge detection).

INDUSTRY EXPERIENCE

Computer Vision Engineer (Intern), CISDI Information Technology — May 2022 – Dec 2023 (3 co-op terms)
Chongqing, China

- Belt-conveyor inspection for steel plants: semantic segmentation (UNet family, Transformer-based) of material type/state and particle size; added GAN refinement of segmentation maps and a Wasserstein-distance test that replaced manual evaluation.
- Independently built a YOLOv5s equipment detector that supplies control signals for automatic converter tapping; sped up TensorRT inference by **~40%** and ported an algorithm to C++ for DeepStream on Jetson Xavier NX / AGX Orin.
- Ported two inspection pipelines to Huawei Ascend (Atlas 200 DK / MindSpore); built a team-wide annotation tool and led 3 interns labelling **~100k** images.

TECHNICAL SKILLS

Programming	C++ (proficient), Python (proficient), MATLAB, C
Robotics / Sim	ROS1 & ROS2, NVIDIA Isaac Sim, Spot SDK, LiDAR–camera calibration, frontier exploration, next-best-view / active perception, SLAM, path planning (TSP/LKH)
Perception / ML	PyTorch, TensorRT, YOLOv7, SAM2, VLM APIs, CNN/deep learning, point-cloud processing (ICP)
Tools	Linux, Docker, Git, SolidWorks, COMSOL

HONORS & TEACHING

- **National Bronze Prize**, 8th China Int'l "Internet+" Innovation & Entrepreneurship Competition, 2023.
- Teaching Assistant — Engineering Design Thinking II (UC, 2025) and C++ Programming (CQU, 2024).